



June 17, 2026

Dear Admission Committee,

I am writing to express my strong interest in pursuing a PhD at the University of Southampton under the title "On-chip quantum cryptography using two-dimensional materials". As an AI and Data Science Engineer with a passion for exploring the intersection of physics, electronics, and materials science, I believe this project offers a unique opportunity for me to contribute to the development of cutting-edge technology.

Throughout my academic career, I have been fascinated by the potential of artificial intelligence to revolutionize various fields. One of my most notable achievements is the development of *MedicalMambaTS*, a linear-time state space model that enables sub-millisecond inference in cardiovascular latency processing and robustness against noise and errors.

Furthermore, my work on *PIDeepONet*, a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd, is a chip-entangled photon source that can be integrated within a quantum cryptography system.

I am particularly drawn to this project because of its potential to enable ultimate security of digital communication based on the laws of quantum physics. The urgency to find alternative techniques to encrypt our data, as highlighted by IBM's recent release of a 1000-qubit quantum computer, is evident. I believe that my skills and experience make me an ideal candidate for this project, and I am excited about the opportunity to contribute to the development of on-chip entangled photon sources.

Thank you for considering my application. I look forward to discussing my qualifications further.

Sincerely,

Ismail Aissaoui